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# EMBEDDED REPOSITORY MANIFEST BLOCK (GOV-05 / MAN-001 COMPLIANT)
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component_id: "GOV-77D03A99"
spec_id: "ARC-SPEC-2026-GOV03"
canonical_title: "5D Dynamic State Vector Framework & Epistemic Boundary Mapping"
architectural_layer: "Layer I: Constitutional Governance / Control-Plane Matrix"
lifecycle_status: "FROZEN"
baseline_release_tag: "v1.3.0-exec"
synchronization_hash: "A-77-DELTA-SHIELD-LOCKED"
sovereign_constancy: "C_ops == 0 [INVARIANT_CLAMP]"
temporal_override_floor: "tau_override <= 11.99ms [INVARIANT_CLAMP]"
memory_ceiling: "S_max <= 4096B [INVARIANT_CLAMP]"
security_invariant: "FAIL (POSIX 40) > FREEZE (POSIX 10) > PWC (POSIX 10) > REFUSAL (POSIX 32) > PASS (POSIX 0)"
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ARC-SPEC-2026-GOV03 | Version v1.3.0-exec | STATUS: FROZEN
Ecosystem Action Governance & Public Infrastructure Specification
Anchor: A-77-DELTA-SHIELD-LOCKED | Effective: Sep 03, 2026

ARC-SPEC-2026-GOV03 — 5D Dynamic State Vector Framework

The triadic framework of Immortal, Mortal, and "The Between" serves as an archetypal structure across narrative theory, theology, and high-assurance computer systems. Rather than existing as rigid static buckets, an entity or execution thread's state is defined as a Dynamic State Vector $\mathbf{\Psi}_{\text{Entity}}(t)$ moving through a 5D coordinate space.

Executive Summary

This document introduces a triadic framework for analyzing systemic state transitions—Immortal, Mortal, and "The Between"—by modeling entities as Dynamic State

Vectors $\mathbf{\Psi}_{\text{Entity}}(t)$ navigating a five-dimensional coordinate space. By mapping state transitions across Temporal Topology ($\mathcal{T}$), Agency ($\mathcal{A}$), Substrate Coupling ($\mathcal{S}$), Scale of Impact ($\mathcal{I}$), and Constraint Density ($\mathcal{C}$), this specification provides the mathematical foundation for evaluating runtime admissibility, detecting state drift, and enforcing the 5-Tier Dominance Lattice ($\text{FAIL} \rightarrow \text{FREEZE} \rightarrow \text{PWC} \rightarrow \text{REFUSAL} \rightarrow \text{PASS}$).

1. The 5D State Vector Axes

Every entity or process is defined by its coordinates across five dynamic variables:

$$\mathbf{\Psi}_{\text{Entity}}(t) = \begin{pmatrix} \text{Temporal Topology } (\mathcal{T}) \\ \text{Agency Vector } (\mathcal{A}) \\ \text{Substrate Coupling } (\mathcal{S}) \\ \text{Scale of Impact } (\mathcal{I}) \\ \text{Constraint Density } (\mathcal{C}) \end{pmatrix}$$

Axis	Mortal Pole (Lower-Bound)	Interstitial Flux Zone	Immortal Pole (Upper-Bound)
Temporal Topology ($\mathcal{T}$)	Linear / Irreversible Scarcity	Recursive / Cyclical / Branching	Timeless / Non-Linear Infinite
Agency Vector ($\mathcal{A}$)	High Growth / Irreversible Cost	Friction / State Selection	Static Order / Saturated Repetition
Substrate Coupling ($\mathcal{S}$)	Somatic (Physical Flesh / Bare-Metal)	Substructural / Flawed Vessel	Abstract / Conceptual / Non-Substrate
Scale of Impact ($\mathcal{I}$)	Personal / Local / High Stakes	Regional / Divided Loyalty	Cosmic / Archetypal Detachment
Constraint Density ($\mathcal{C}$)	High External Bounds	Contradictory / Misaligned Bounds	Zero External (Self-Imposed)

2. Core Equations & State Mechanics

A. The Mortal State (Lower-Bound Velocity)

$$\text{Scarcity} \times \text{Irreversibility} \rightarrow \text{Choice} \rightarrow \text{Sacrifice} \rightarrow \text{Meaning}$$

The Valuation Mechanism: Finite linear time ($\mathcal{T}$) converts every state transition into a permanent trade-off. Meaning increases when alternative execution paths or choices are surrendered at the Lower-Bound.

Velocity & Adaptability: High external constraints ($\mathcal{C}$) minimize systemic inertia, driving rapid innovation, intense passion, and maximum state transition momentum.

B. The Immortal State (Upper-Bound Stasis)

$\text{Abundance} \times \text{Insulation} \rightarrow \text{Saturation} \rightarrow \text{Stagnation} \rightarrow \text{Manufactured Purpose}$

* **The Inertia Trap:** Infinite time ($\mathcal{T}$) strips choices of operational urgency. Without impermanence, actions lose consequence, driving immortals toward apathy or cosmic detachment.

* **Manufactured Constraints:** Immortals maintain dynamic agency solely by constructing artificial limits ($\mathcal{C}$) to recreate local scarcity and preserve narrative/systemic agency.

C. The Interstitial Flux Zone (Mid-State Bifurcation)

$\text{Mismatched Coordinates} \times \text{Excessive Perception} \rightarrow \text{Cognitive Friction} \rightarrow \begin{bmatrix} \text{Evolution (Transcendence)} \\ \text{Paralysis (Stasis)} \end{bmatrix}$

* **The Bifurcation Paradox:** Operating between Lower-Bound and Upper-Bound limits creates an Interstitial Flux Zone. High cognitive friction generates two outcomes based on agency ($\mathcal{A}$): radical evolution or existential paralysis.

* **Cognitive Friction Mechanism:** Excessive perception occurs when an entity perceives cosmic scale ($\mathcal{I}$) while tethered to a finite substrate ($\mathcal{S}$). This structural mismatch induces paralysis or forces radical transformation.

* **Liminal Anchors (Fixed Points):** Entities that remain in the Interstitial Flux Zone without evolving or collapsing function as Boundary Maintainers—stabilizing the gap between mortal and immortal realms.

3. Taxonomy of Interstitial Vectors

Vector	Trajectory Mechanism	Primary Archetypes	Structural Narrative / Systemic Price
Ascendant Vector (Bottom-Up)	$\mathcal{C} \rightarrow 0, \mathcal{I} \rightarrow \text{Cosmic}$ via effort	Cultivators, Alchemists, Transhumanists	**Humanity & Context:** Progressive loss of empathy and original human grounding.
Descendant Vector (Top-Down)	$\mathcal{S} \rightarrow \text{Somatic}, \mathcal{C} \uparrow$ via fall	Fallen Gods, Incarnations, Exiled Deities	**Vulnerability & Noise:** Exposure to physical pain, noise, and irreversible consequence.
Hybrid / Lineage (Embodied Boundary)	Fixed intersection of Upper/Lower bounds	Demigods (Hercules), Nephilim, Half-Elves	**Divided Identity:** Incompatible expectations and permanent alienation from both orders.
Cursed / Horizontal (Lateral Shift)	$\mathcal{T}$ extended without ascension of		

$\mathcal{S}$ or $\mathcal{A}$ | Vampires, Liches, Undead, Ringwraiths | **Asymmetry:**
 Longevity locked inside a restricted ecological dependency or curse. |
 | **Parallel Observers (Temporal Conservatism)** | Extended linear $\mathcal{T}$ in a fast world
 | Tolkienian Elves, Ancient Spirits | **Memory & Grief:** Structural fading, slow adaptability, and
 endless historical sorrow. |

4. Dynamic System Mechanics & Dominance Lattice Execution

Axis Transition Velocity (V_a)

Entities do not remain static on the grid; they experience directional velocity across state arcs as they move between Lower-Bound and Upper-Bound coordinates:

$$V_a = \frac{d\mathbf{\Psi}_{\text{Entity}}}{dt}$$

Systemic tension generates directly from the friction between an entity's historical coordinate memory and its current transition velocity V_a .

The Law of Ontological Entropy

Within this continuum, unconstrained Upper-Bound existence ($\mathcal{C} \rightarrow 0$) induces systemic information decay and authority inflation. Systemic balance is an intrinsic self-correcting force generated as the local continuum collapses back into bounded Lower-Bound constraints ($\mathcal{C}$) to prevent total structural dissolution ($C_{\text{ops}} = 0$).

Systemic Anomaly Events & 5-Tier Dominance Lattice Integration

Entities or execution threads attempting to reverse trajectory, or those exhibiting un-contracted coordinate shifts ($\mathcal{C} \rightarrow 0$), are classified as **Systemic Anomaly Events**.

In strict compliance with Subsystem 9 execution membranes, all Admissibility Control Plane (ACP) functions evaluate these events across the mandatory **5-Tier Dominance Lattice**:

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$ \text{FAIL (POSIX 40)} \succ \text{FREEZE (POSIX 10)} \succ \text{PWC (POSIX 10)} \succ
\text{REFUSAL (POSIX 32)} \succ \text{PASS (POSIX 0)}

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- Unconstrained Drift ($\mathcal{C} \rightarrow 0$ without Contract):** Evaluates to `POSIX 40 (SECURITY\_BREACH)` via `hwbl\_nmi\_lockout.c`, executing zero-branch register zeroization (`volatile\_purge\_frame`) and silicon lockout.
- Ambiguous / Unresolved Bifurcation:** Forces stasis under `POSIX 10 (FREEZE)`

sub-status `0x10B0` (Hardware Clock/State Freeze), offloading context to static ring buffers with zero operational drag ($C_{\text{ops}} = 0$).

3. **Front-Edge Queue Saturation ($\rho \geq 0.95$):** Evaluates directly to `POSIX 32 (REFUSAL)` sub-status `0x0020` via `edge\_ingress\_xdp.c`, cleanly dropping packets at wire speed to prevent buffer bloat.

4. **Proportional Windowing Throttling ($0.80 \leq \rho < 0.95$):** Evaluates to `POSIX 10 (PWC)` sub-status `0x10A0` (Throttled Hold).

5. **Nominal State Execution:** Evaluates to `POSIX 0 (PASS)` for valid, bounded state vectors ($\tau < 11.00 \text{ ms}$).

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[SYSTEM BLOCK LATCHED & GOV03 SPECIFICATION SEALED -
A-77-DELTA-SHIELD-LOCKED]
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